



Role of martensite decomposition in tensile properties of selective laser melted Ti-6Al-4V

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ABSTRACT

The as-Selective Laser Melted (as-SLMed) Ti-6Al-4V generally has a martensitic microstructure. In the present study, martensite decomposition was investigated in various annealed samples by XRD, SEM and TEM. It has been found that annealing at 700 °C or 800 °C for 2 h was not sufficient to obtain a fully decomposed microstructure. The amount of martensite did reduce greatly and volume fraction of β phase increased significantly when the annealing temperature increased from 700 to 800 °C. However, the best combination of mechanical property was not obtained. It is to be noted that annealing at 800 °C for 2 h is widely used for SLMed Ti-6Al-4V as reported in literature. Detailed SEM and TEM analysis revealed the presence of residual martensite containing twins and a high density of dislocations even in the 800 °C/2 h annealed sample. The twin was determined as $\{10\bar{1}1\} <10\bar{1}2>$ type, which are retained twin formed in martensitic transformation during the cooling in SLM. Annealing at 800 °C for 6 h or more provided a fine equilibrium $\alpha + \beta$ microstructure, which provided a strong and ductile SLMed Ti-6Al-4V.

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1. Introduction

Titanium (Ti) alloys have a wide range of applications in different industries ranging from chemical to aerospace owing to their high specific strength, superior mechanical property at elevated temperatures, and outstanding corrosion resistance [1–4]. However, the high cost of Ti alloys is a limitation in extending their applications [5]. Conventional processing technique is a contributor to the high cost due to significant material wastage and high energy input [5]. To reduce processing cost, Selective Laser Melting (SLM) has demonstrated its advantage in reducing material wastage as it only requires raw powders for building components of various shapes with limited machining [6]. In addition, un-consumed powders can also be recycled for future production. Therefore,

SLM is a promising materials processing technique to reduce the high cost of producing Ti parts.

Compared to wrought Ti-6Aluminum-4Vanadium (Ti-6Al-4V in wt.%), as-Selective Laser Melted (as-SLMed) Ti-6Al-4V products have generally higher yield strength (YS) and ultimate tensile strength (UTS) but lower ductility. This can be attributed to the fine martensitic (α') microstructure resulted from the fast cooling of the melt pool [7–12], and some levels of thermal stresses built up during the SLM process [13–15]. Therefore, post heat treatment is required to homogenize the microstructure, to transform martensite (α') into equilibrium $\alpha + \beta$ phases, and to minimize thermal stress [16,17].

Previous studies [14,15,17–21] investigated extensively the influence of post heat treatment on retained thermal stress, microstructure evolution and resultant mechanical property in SLMed Ti-6Al-4V. There are generally three types of heat treatments according to recent articles in literature [22,23] - stress relief (480–650 °C), annealing (up to 870 °C), and a two stages treatment of solution treatment (955–970 °C) followed by ageing (480–595 °C) [22]. For stress relieve efficiency, a heat treatment on Ti-6Al-4V at 595 °C for 1 h (h) decreased the residual stress from

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approximate 725 MPa to 50 MPa [24]. Another two studies reported that 2 h at 730 °C was sufficient to eliminate the residual stress of SLM Ti-6Al-4V parts [17,18]. Comparing to stress relieve treatments, annealing is generally carried out at relatively higher temperatures up to 870 °C. Temperatures in the range of 600–850 °C were found to be effective in promoting the martensite decomposition [20]. Another study [25] also revealed that martensite decomposition can be easily observed for heat treatment conducted in the temperature range of 750–900 °C. Hence, annealing has a dual effect on SLMed Ti-6Al-4V: (i) reducing the residual stress, and (ii) promoting martensite decomposition.

To enhance the ductility, the complete decomposition of brittle martensite is critical. As shown in Table 1 [16–19,26–29], the annealing treated SLMed Ti-6Al-4V has a strength comparable to wrought Ti-6Al-4V, but its ductility is lower than that of wrought Ti-6Al-4V. The highest elongation of the heat treated SLM Ti-6Al-4V is 12.8% in Table 1. In contrast, hot worked and annealed Ti-6Al-4V has a much higher elongation at 18.1% [14]. Table 1 also shows that a 'complete' martensite decomposition (resultant microstructure is $\alpha + \beta$) can only be achieved at annealing temperatures above 780 °C for 2 h. However, these studies only employed characterization techniques including XRD and SEM, which may not be able to detect the existence of a small amount of retained martensite at sub-micron size. It is likely that the martensite decomposition is not complete in these works. Therefore, more investigations need to be conducted on the completeness of martensite decomposition in annealed SLM Ti-6Al-4V, as it can strongly influence the resulting ductility.

This study focuses on optimizing the annealing treatment for SLMed Ti-6Al-4V to improve the ductility without sacrificing the strength. Experiments were systematically conducted on the influences of annealing temperatures and time on martensite decomposition, microstructure evolution, and resultant mechanical property by XRD, SEM, and TEM. Detailed TEM experiment showed that an equilibrium $\alpha + \beta$ microstructure was achieved after specific annealing condition shown in this work, indicating a complete decomposition of martensite. Such SLMed and annealed Ti-6Al-4V has ductility (~18%) similar to wrought Ti-6Al-4V as well as more than adequate strength.

2. Material and methods

Gas atomized pre-alloyed Ti-6Al-4V powder was provided by Falcon Tech Co., Ltd. As shown in Fig. 1, most powders were spherical, and the insert shows that some powders have fine satellites attached with size below 5 μm . The D(10), D(50) and D(90) of

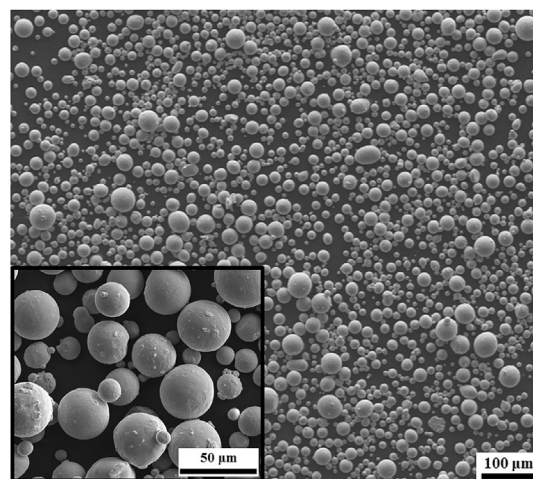


Fig. 1. Secondary electron SEM image of gas atomized Ti-6Al-4V powders with a diameter up to 63 μm , and the insert is a high magnification SEM image of Ti-6Al-4V powder.

powders were 16.2 μm , 29.7 μm and 50.1 μm , respectively. Powder composition (wt.%) is 6.14Al-4.13V-0.008C-0.03N-0.001H-0.09Fe-bal.Ti. All SLMed samples were built by an EOS M280 machine under a 99.99% argon (Ar) atmosphere. During the deposition, the Ti-6Al-4V substrate was pre-heated to 100 °C, and the laser spot size was 0.1 mm. The laser power and scan speed were 280 W and 1200 mm/s; the layer thickness and hatch distance were 30 μm and 140 μm respectively. For each layer, the laser scan direction was altered by 66.67° from its previous layer. Oxygen (O) concentrations for the Ti-6Al-4V powder before deposition and annealed SLM samples were determined to be 1550 ppm and 1715 ppm respectively by Luvak Inc. (MA, United States) through an inert gas fusion method. Oxygen pick-up was 165 ppm during SLM and the following vacuum annealing. Density of SLMed Ti-6Al-4V was measured at 99.9% by image analysis of five optical micrographs.

According to Table 1, 800 °C for 2 h is a popular anneal treatment for SLMed Ti-6Al-4V. Comparing to 800 °C for 2 h, 700 °C for 2 h was carried out to study the effect of annealing temperature, and 800 °C for 6 h and 12 h were conducted to investigate the effect of annealing time. All samples were annealed in a Nadcap certified vacuum furnace. After annealing, Furnace Cool (FC) was carried out by turning the furnace off, and the cooling rate was in the range of 2–3 °C/minutes (min).

Cuboid samples (10mm×10 mm×100 mm) were horizontally

Table 1
SLM parameters, microstructure and mechanical properties of SLM Ti-6Al-4V with subsequent annealing heat treatment. Air Cool (AC) and Furnace Cool (FC) are cooling methods after annealing. Laser power, P ; layer thickness, d ; scan speed, v ; hatch distance h ; Yield Strength, YS; Ultimate Tensile Strength, UTS; Elongation, El.

	P (W)	d (mm)	v (mm/s)	h (mm)	Spot (mm)	Pre-heating (°C)	Annealing and cooling	Microstructure	YS	UTS	El. (%)
SLM [26]	150–200	0.02	800–1500	0.075	0.15	—	700 °C/2 h + FC	$\alpha + \beta + \alpha'$	—	—	—
SLM [27]	200	0.04	1250	—	0.15	—	700 °C/2 h + 10 °C/min cooling	α'	1051 (V)	1115 (V)	11.3 (V)
SLM [17]	160	0.04	220	0.2	0.22	500	730 °C/2 h + AC	$\alpha + \beta + \alpha'$	965 ± 16 (H) 900 ± 101 (V)	1046 ± 6 (H) 1000 ± 53 (V)	9.5 ± 1 (H) 1.9 ± 0.8 (V)
SLM [18]	157	0.05	225	0.1	0.07	—	730 °C/2 h + FC	$\alpha + \beta$	958 ± 6 (H) 937 ± 9 (V)	1057 ± 8 (H) 1052 ± 11 (V)	12.4 ± 0.7 (H) 9.6 ± 0.9 (V)
SLM [16]	250	0.03	1600	0.06	0.052	—	780 °C/2 h + FC (~2.4 °C/min)	$\alpha + \beta$	—	—	—
SLM [19]	—	0.03	—	—	—	100	800 °C/2 h + FC	$\alpha + \beta$	962 (V)	1040 (V)	5 (V)
SLM [28]	—	0.03	—	—	—	100	800 °C/2 h	$\alpha + \beta$	—	1228.1 ± 32.4 (V)	8 ± 1.5 (V)
SLM [29]	170	0.03	1250	0.1	—	—	800 °C/2 h + AC	$\alpha + \beta$	—	—	—
SLM [16]	250	0.03	1600	0.06	0.052	—	850 °C/2 h + FC	$\alpha + \beta$	955 ± 6 (H)	1004 ± 6 (H)	12.84 ± 1.36 (H)

deposited on the substrate with the long axis parallel to the substrate. After anneal, these horizontal samples were cut off from the substrate by EDM, and machined into cylindrical tensile specimens (gage diameter at 4 mm and gauge length at 16 mm) according to ASTM E8/E8M standard [30]. Mechanical properties of Yield Strength (YS), Ultimate Tensile Strength (UTS) and elongation (El.) were obtained by using these cylindrical specimens, and an extensometer was used to ensure an accurate strain measurement. Three replicates of horizontally built samples for each condition were tested, where the tensile axis was perpendicular to the build (Z) direction.

Microstructure characterization was conducted by XRD, SEM and TEM on horizontal cross sections. For XRD and SEM, all the samples were ground down to a finish by 4000-P SiC abrasive papers, and then were polished with a mixed solution of 15 vol% H₂O₂ and 85 vol% Oxide Polishing Suspension (OP-S) on a MD-Chem polishing plate. XRD was performed at 40 kV, 40 mA, and a step size of 0.005° at room temperature by a Bruker D8 Advance diffractometer (Cu-K_α) equipped with a Lynxeye XE detector. Secondary Electron (SE) and Back-scattered Electron (BSE) images were obtained on a FEI Quanta 3D FEG microscope. Based on SEM images, lamellar thickness (5 image analyses for each sample condition) and volume fraction (30 image analyses for each sample condition) measurements were carried out by ImageJ software. Fig. S1 in Supplementary material is an example of volume fraction measurement. For TEM, 120 μm thin foils were prepared by mechanical grinding, and then they were electropolished by a Tenupol twin jet polishing system in a solution of 57 vol% methanol, 40 vol% n-butanol, and 3 vol% perchloric acid at 40 V and −35 °C. Selective Area Diffraction (SAD) patterns and Bright Field (BF) TEM images were obtained on a FEI Tecnai G2 T20 TWIN TEM at 200 kV. All BF images were taken at about 3–5° away from the zone axis to increase the contrast.

3. Results

3.1. Microstructure

Fig. 2a shows XRD results of as-SLMed Ti-6Al-4V and various annealed conditions. The vertical dashed lines locate the peak positions in the as-SLMed sample. The as-SLMed condition has a relatively broader peak width comparing to other annealed conditions, which is caused by the presence of α' martensite. Because martensite is supersaturated in V content, and higher V content results in a greater elastic distortion of HCP crystal, which broadens α' peaks [29,31,32]. Peak becomes narrow and sharp after annealing, which indicates the starting of martensite decomposition (α' → α + β). After annealing at a higher temperature of 800 °C for longer time, the peaks started to move towards the high angle. This could be attributed to solute redistribution after prolonged anneal at 800 °C [26]. Solute diffusion introduces increased concentrations of Al in HCP α phase and V in BCC β phase. Both Al (0.143 nm) [33] and V (0.132 nm) have smaller atomic radius than Ti (0.147 nm) [29], therefore the solute redistribution decreases the lattice parameters and results in peak shift towards higher angles. Using logarithm axis (Fig. 2b) revealed the β {110} peak in the higher annealing temperature situations at 800 °C for 2 h, 6 h, and 12 h, which was not observed in as-SLMed and low temperature (700 °C) annealed samples.

Fig. 3 presents BSE micrographs of as-SLMed and annealed samples. For the as-SLMed sample, the microstructure is α' martensite containing twins (Fig. 3a). The decomposition of α' martensite into α and β phases occurs with annealing. After annealing at 700 °C for 2 h (Fig. 3b), a small volume of fine β layers and particles appeared, which was not displayed in XRD (Fig. 2) due

to the detection limit. The size of α lamellae and volume fraction of β increased at higher annealing temperature (800 °C) compared to those of the 700 °C. Interestingly, twins are still observed after the 2 h annealed conditions at 700 °C and 800 °C (white arrows and associated inserts in Fig. 3b and c). At 800 °C, the effect of annealing time is shown in Fig. 3c (2 h), d (6 h), and e (12 h). The size of lamellae is slightly increased after prolonged annealing, and the amount of β phase increases with annealing time from 2 h to 6 h. After which the amount of beta phase remains relatively constant.

TEM was further used to characterise the twin structures observed in the 2 h annealed samples. Fig. 4a and b shows twins and a high density of dislocations after annealing at 700 °C and 800 °C for 2 h. According to the SAD patterns, they are identified as {10 $\bar{1}$ 1} type twins. As shown by a lower magnification image in Fig. 4c, these twins disappeared, and a recovered microstructure was obtained in the annealed condition of 800 °C for 6 h.

After detailed image analysis, the measured width of α lamellae and volume fraction of β phase are described in Fig. 6a and Fig. 7 respectively. In the annealed samples, α lamellae have a width of approximately 0.68 μm (700 °C for 2 h), 0.87 μm (800 °C for 2 h), 1.03 μm (800 °C for 6 h), and 1.11 μm (800 °C for 12 h). β volume fraction increased with annealing temperatures and time from 1.50% in 700 °C for 2 h to 4.99% in 800 °C for 2 h, and finally settled at 6.15% in 800 °C for 6 h and 12 h.

3.2. Tensile properties

Fig. 5 shows the tensile property of as-SLMed and all four annealed conditions. As-SLMed condition has a martensitic microstructure, which gives high YS and UTS, but low ductility. After annealing, YS and UTS generally follow a similar trend. They decrease with increasing annealing temperatures and time while ductility has an opposite trend. It can be seen that the lower annealing temperature of 700 °C resulted in high YS of 1011 MPa and low El. of 13.5%. Once a higher annealing temperature of 800 °C for 2 h was applied, there was a small increase in elongation to 14.7% and a large reduction of YS to 953 MPa. Further annealing at 800 °C to 6 h increased the ductility significantly to 18.9% and maintained a high elongation larger than 18% for 12 h sample. Such a gain in ductility came with only a 2% decreases in YS after further annealing. The YS was approximate 930 MPa after 12 h annealing at 800 °C.

4. Discussion

4.1. Martensite decomposition in annealing

α' martensite are formed in the as-fabricated Ti-6Al-4V due to rapid cooling in SLM [7,13,16–19,26,34]. Such a microstructure favours intergranular failure inside columnar β grains and results in a reduced ductility [7,15]. The martensitic transformation is a shear type process, which transforms the Body-Centered Cubic (BCC) β into hexagonal α' [1]. As martensitic transformation is diffusionless, α' is supersaturated with β stabilizers. Martensite decomposition starts once annealing is in the α + β phase field, and β particles form at dislocations and/or β layers forms at martensite boundaries [1].

Fig. 7 describes the effect of annealing on the volume fraction of β phase. It is worth reminding the reader that these samples were furnace cooled at 2–3 °C/min. The volume fraction of β rose with increased annealing temperature and time till the 800 °C for 6 h condition. After 6 h, the β volume fraction stabilized at 6.15%. This indicated that the diffusion controlled martensite decomposition, α' → α + β, did not reach the equilibrium at 800 °C for 2 h. In addition, there is still some retained martensite thereby suggesting a need for longer annealing time (>2 h) for all martensite to be

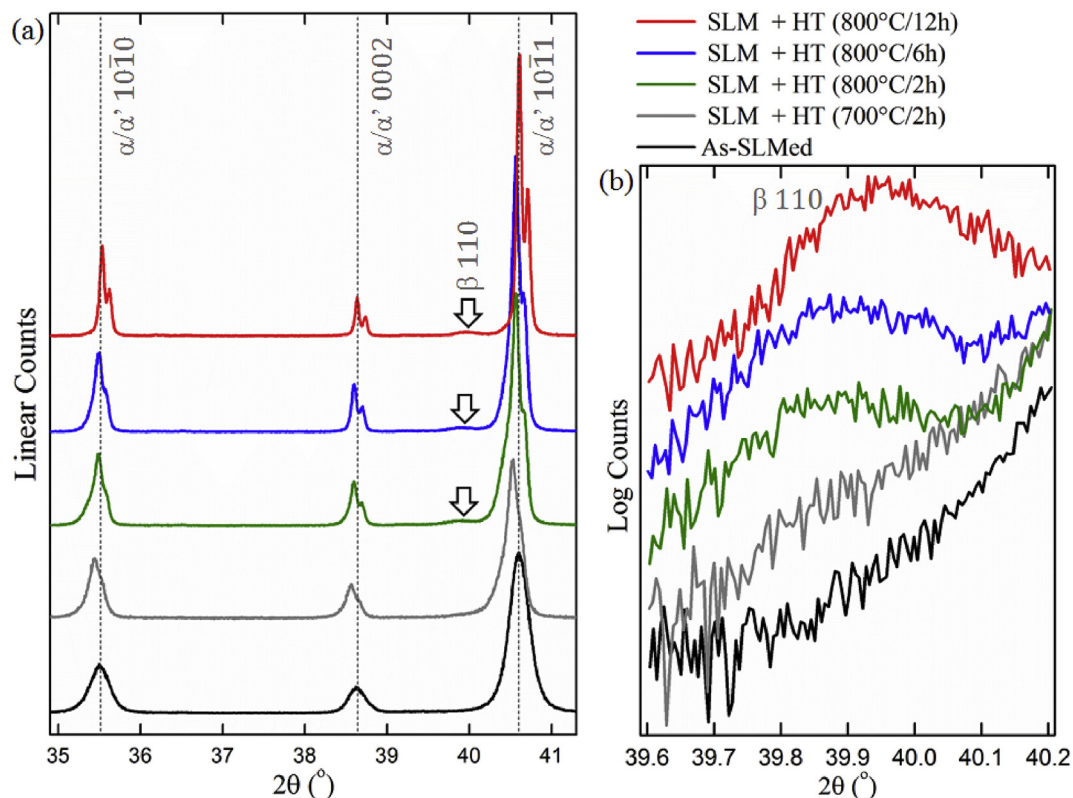


Fig. 2. (a) XRD spectra of as-SLMed and annealed samples; vertical dashed lines denote the peak positions in as-SLMed Ti-6Al-4V, and (b) XRD spectra with log scale of counts show β {110} peak around 39.9° .

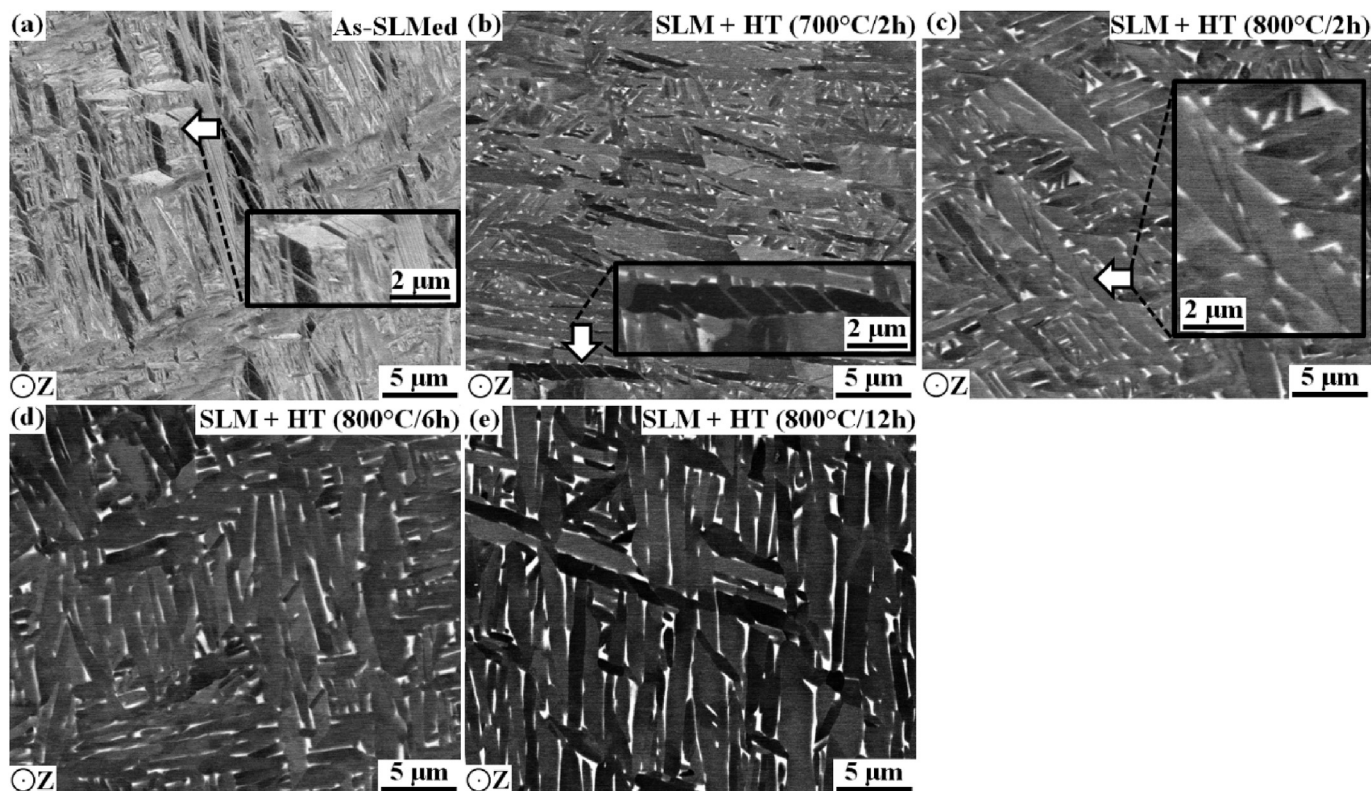


Fig. 3. BSE images of (a) as-SLMed Ti-6Al-4V; and various post-SLM annealed conditions at (b) 700°C for 2 h, (c) 800°C for 2 h, (d) 800°C for 6 h, and (e) 800°C for 12 h. White arrows denote twins inside lamella, and associated inserts are high magnification BSE images of twins. In (b)–(e), α is the dark phase and β is the white bright phase.

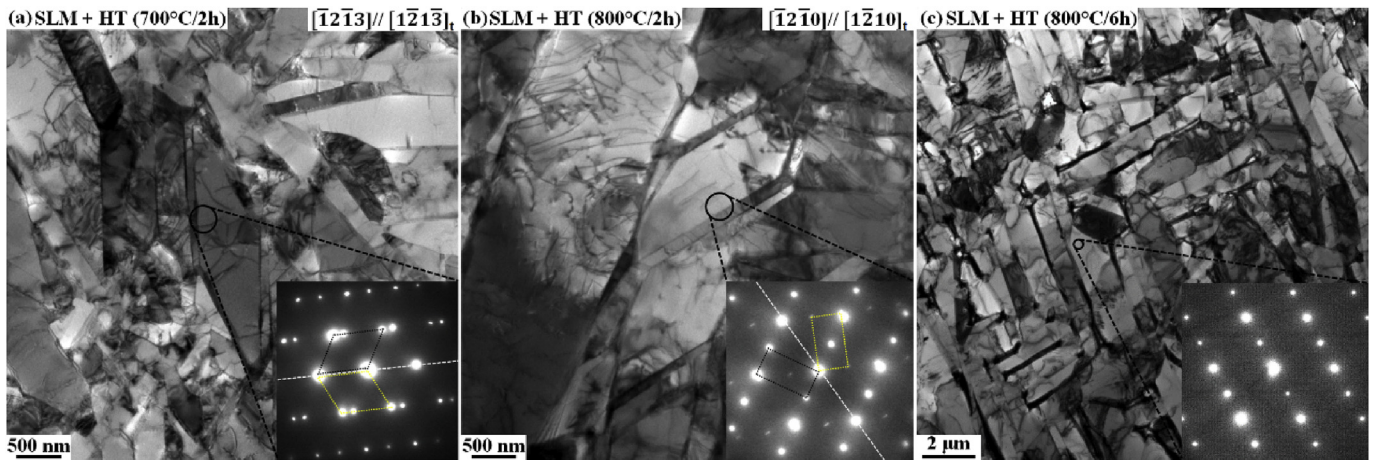


Fig. 4. Bright Field (BF) images of various post-SLM annealed conditions at (a) 700 °C for 2 h, $B \approx [1\bar{2}1\bar{3}]/[1\bar{2}1\bar{3}]_t$; (b) 800 °C for 2 h, $B \approx [1\bar{2}1\bar{0}]/[1\bar{2}1\bar{0}]_t$; and (c) 800 °C for 6 h, $B \approx [1\bar{2}1\bar{0}]$. All the black circles show the locations of selective area aperture for diffraction, and the white dashed lines illustrate the twin symmetry.

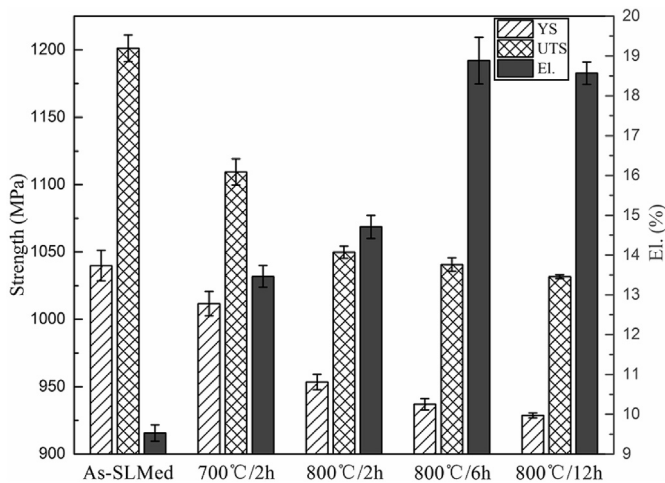


Fig. 5. Tensile properties of as-SLMed and various annealed horizontally built samples, which has the tensile axis perpendicular to the SLM build direction.

decomposed. In a study of wrought Ti-6Al-4V heat treated at various temperatures followed by water quench [35], β volume fraction was determined by electro microprobe and BSE image

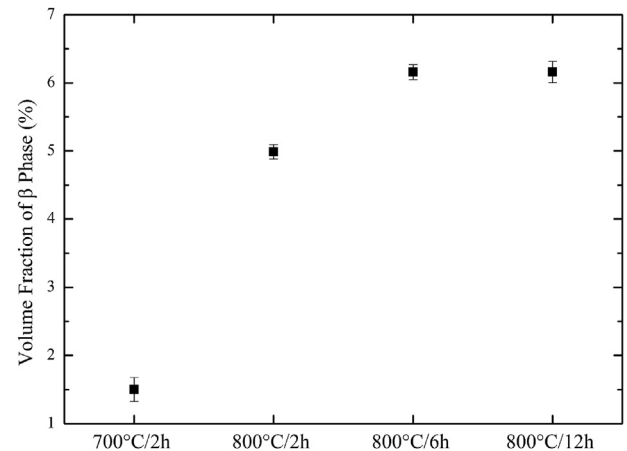


Fig. 7. Measured volume fraction of β phase on horizontal plane by image analysis.

analysis. These two methods had a consistent measurement, which showed there was about 5% difference in volume fraction of β phase between 700 °C and 800 °C [35]. The FC condition was used in the current study to allow for a near-equilibrium cooling. During the slow cooling, a certain volume fraction of β (formed at 700 and/or

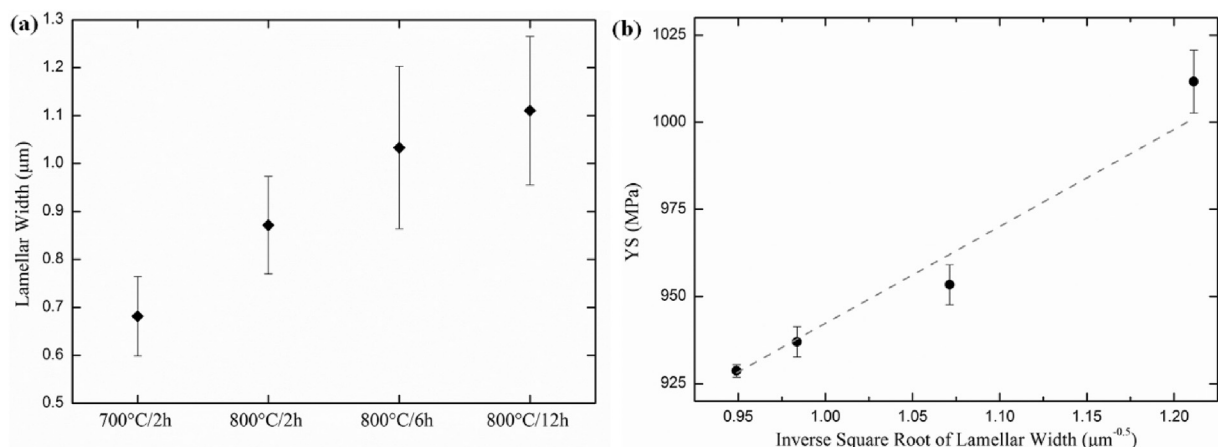


Fig. 6. (a) Measured α lath width on horizontal plane by image analysis; and (b) a Hall-Petch relationship between yield strength and measured α lath width in (a).

800 °C) transforms to the low temperature stable α phase [17]. The difference in volume fraction of β between 700 °C/2 h and 800 °C/6 h heat treated samples in our study was close to 5%. This result agrees with the literature result of wrought Ti-6Al-4V [35].

Martensite generally contains a high density of dislocations and some twins [1,36]. Previous SLM Ti-6Al-4V investigations [7,37,38] reported $\{10\bar{1}2\}$ type twinning system. Its activation was suggested to accommodate the thermomechanical stress in high tensile residual stressed regions, such as on the top and in the volume of SLM built samples [37]. In the present study, $\{10\bar{1}1\} <10\bar{1}2>$ type twin and high density of dislocations were found in the 2 h annealed samples at 700 °C and 800 °C (Fig. 4a and b). To the authors' knowledge, this is the first $\{10\bar{1}1\}$ twin reported for SLMed Ti-6Al-4V. Twins were not observed in the sample that went through prolonged annealing (6 h) at 800 °C, which had a recovered lamellar microstructure as shown in Fig. 4c. This indicates that the heat treatment was adequate in removing dislocations associated with the martensite formation.

$\{10\bar{1}1\}$ twin was reported in martensitic transformation $\beta \rightarrow \alpha'$ in various Ti alloys including Ti-6wt.%Al-4wt.%V [39], Ti-8wt.%V [40], Ti-8wt.%V-4wt.%Sn [40], Ti-6wt.%Cu [41], and Ti-5.1 at.%Cr [42]. Normally twinning as a deformation mode only occurs in Commercial Pure (CP) Ti at high temperatures, and suppressed by adding alloy elements of Al and O [1]. However, it is known that $\{10\bar{1}1\}$ twin is accompanied with martensitic transformation. Therefore, the observed $\{10\bar{1}1\}$ twin in the present work should be retained twin formed during martensitic transformation during the cooling of SLM. In addition to the volume fraction of β phase discussed before, such retained twin in martensite is a further evidence suggesting that the diffusion controlled martensite decomposition is not complete at the 800 °C for 2 h condition.

4.2. Tensile properties

4.2.1. Strength

As-fabricated Ti-6Al-4V contains a martensitic microstructure, which leads to low ductility [7,8,10]. Upon annealing, α' martensite starts to transform to $\alpha + \beta$, and the lamellae become thicker. The resultant mechanical properties are reduced strength (both YS and UTS) and increased ductility [43].

Fig. 6a summarizes the measured width of lamellae on the horizontal plane. For 2 h annealed treatment, the lath thickness increased about 190 nm for the 800 °C annealed sample compared to the 700 °C. For prolonged annealing at 800 °C for 12 h, α lath only increased a further 130 nm after an additional 6 h holding time compared to that of 800 °C for 6 h. The coarsening kinetics are controlled by the diffusion of V in Ti-6Al-4V [44], which explains the annealing temperature having a more significant effect in lamellar growth compared to time when heat treating well below β transus. Fig. 6b plots the YS with the inverse square root of α lamellae width. It reveals that the YS has a Hall-Petch type relationship with the lath thickness, and similar observations has been reported in other studies [20,45]. It is worth noting that even after a long exposure (6 h) at 800 °C, a strong annealed SLM Ti-6Al-4V is obtained through a fine lamellar microstructure ($\sim 1 \mu\text{m}$) having YS at ~ 937 MPa and UTS at ~ 1041 MPa.

4.2.2. Ductility

As discussed in section 4.1, 2 h annealing at both 700 °C and 800 °C resulted in a lamellae microstructure with retained martensite containing twins and a high density of dislocations. Retained martensite is brittle, and results in inadequate ductility and the presence of large amount of dislocations around them are locations for high stress concentration and void formation. In addition to the TEM evidence, measured β volume fraction (Fig. 7)

proves that the martensite decomposition was not complete at 800 °C for 2 h. On the other hand, the annealing at 800 °C for 6 h (Fig. 4c) shows a fully recovered $\alpha + \beta$ equilibrium microstructure. Detailed electron microscopy also found no retained twins in martensite for the samples annealed at 800 °C for 6 h. This explains why the annealed samples at 800 °C for 6 h and 12 h have a much higher elongation ($>18\%$) with respect to that of the 2 h annealed situations i.e. both 700 and 800 °C. After a complete decomposition of α' into equilibrium $\alpha + \beta$ lamellae (eg. 6 h at 800 °C), the ductility of SLM and annealed Ti-6Al-4V ($>18\%$) is comparable to the wrought Ti-6Al-4V [14] at 18.1%.

5. Conclusions

Various annealing treatments were performed on as-SLMed martensitic Ti-6Al-4V samples to study the effects of thickness of α lath, volume fraction of β phase, and retained martensite on mechanical property. The following conclusions were reached.

- Yield strength is governed by α lath thickness through Hall-Petch relationship in annealed samples. Prolonged annealing at a temperature well below β transus only slightly coarsens the microstructure, therefore the reduction in strength is minimal during annealing.
- The retained twin is identified as $\{10\bar{1}1\} <10\bar{1}2>$ type, which should be formed by martensitic transformation during the cooling of SLM.
- Measured volume fraction of β phase and TEM evidence of retained twins in martensite suggest that 2 h at both 700 °C and 800 °C are insufficient for a complete martensite decomposition. Retained martensite is brittle, and results in inadequate elongation.
- An annealing of 6 h at 800 °C leads to a fully decomposed equilibrium $\alpha + \beta$ lamellar microstructure, which provides a strong and ductile SLMed Ti-6Al-4V with ductility similar to wrought Ti-6Al-4V.

Conflicts of interest

The authors have no competing interests to declare.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.jallcom.2018.02.111>.

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